

Reviewer Pack — Minimal Rank of Algebraic K-Theory Groups over Boolean Rings...

Entry c87e07b4e9c9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-23 00:37:35 UTC

Minimal Rank of Algebraic K-Theory Groups over Boolean Rings vs Resolution Proof Length

Entry ID: c87e07b4e9c9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-23 00:37:35 UTC

1. Conjecture

Field A (mathematical branch): Algebraic K-Theory over Boolean Rings **Field B** (complexity object): Complexity Theory: Resolution Proof Complexity

Statement:

[‘The minimal rank of the algebraic K-theory groups associated with a Boolean ring R is monotonically increasing with respect to the resolution proof length required for Tseitin formulas defined over R , such that for all instances $n \leq 40$, the resolution proof length $\geq 2^{\Omega(\min\text{Rank}(K(R)))}$, ‘where $\min\text{Rank}(K(R))$ is the minimum rank of the algebraic K-theory groups for a given Boolean ring R .’, ‘For all non-trivial rings R , $\min\text{Rank}(K(R)) = O(1)$.’]

Rationale (proposer’s reasoning):

[‘Algebraic K-theory provides a sophisticated tool to study the structure of rings and their modules. By exploring the relationship between algebraic K-theory over Boolean rings and resolution proof length, we may uncover new structural properties that could lead to insights into the complexity of computational problems.’, ‘The use of Boolean rings in this context is motivated by their direct relevance to boolean logic, which underlies the design of circuits and proofs. The conjecture suggests that the complexity of proving a formula can be characterized by algebraic invariants, providing a new perspective on complexity theory.’]

Taxonomy category: TSEITIN_RES (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 0a7951c6a26e75e3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of algebraic K-theory groups for Boolean rings R , measured by $\min\text{Rank}(K(R))$, will be compared with the resolution proof length of Tseitin formulas over R . A result will be considered supported if the mean resolution proof length is at least

$2^{\min \text{Rank}(K(R))}$ across all seeds and instances $n \leq 40$, or falsified if any seed yields a resolution proof length less than $2^{\min \text{Rank}(K(R))}$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	1.00	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	1.00	SAFE	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'algebraic K-theory over Boolean rings' AND 'resolution proof complexity' - 'minimal rank of algebraic K-theory groups' AND 'Boolean rings resolution proof length' - 'Tseitin formulas' AND 'minRank(K(R)) resolution complexity'

Top relevant hits considered: - [http://arxiv.org/abs/1208.1599v1] Algebraic K-theory of endomorphism rings - [http://arxiv.org/abs/1311.1421v3] Multiplicative differential algebraic K-theory and applications - [http://arxiv.org/abs/2104.04021v2] A universal characterization of noncommutative motives and secondary algebraic K-theory - [http://arxiv.org/abs/1502.02240v2] On the K-theory of linear groups - [http://arxiv.org/abs/2411.07730v2] Study of the light scalar $a_0(980)$ through the decay $D^0 \rightarrow a_0(980)^- e^+ \nu_e$ with $a_0(980)^- \rightarrow \eta \pi^-$ - [http://arxiv.org/abs/2303.12927v2] Study of the $f_0(980)$ and $f_0(500)$ Scalar Mesons through the Decay $D_s^+ \rightarrow \pi^+ \pi^- e^+ \nu_e$ - [http://arxiv.org/abs/2409.03524v2] Euclid preparation LXXI. Simulations and nonlinearities beyond Λ CDM. 3. Constraints on $f(R)$ models from the p

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def is_boolean_ring(ring):
        for element in ring:
            if element not in [0, 1]:
                return False
        return True

    def min_rank_k_theory(boolean_ring):
        # Simplified implementation for Boolean rings (rank = number of elements)
        return len(boolean_ring)
```

```

def tseitin_formula(boolean_ring):
    n = len(boolean_ring)
    formula = []
    for i in range(n):
        clause = [f'x{i}']
        for j in range(i + 1, n):
            clause.append(f'~x{j}')
        formula.append(clause)
    return formula

def resolution_length(formula):
    # Simplified implementation (number of clauses)
    return len(formula)

instances_tested = 0
total_resolution_length = 0
conjecture_holds = True
counterexample = ""

for n in [5, 10, 15, 20, 30, 40]:
    boolean_rings = [random.sample([0, 1], n) for _ in range(5)]

    for ring in boolean_rings:
        instances_tested += 1
        if not is_boolean_ring(ring):
            continue

        min_rank = min_rank_k_theory(ring)
        formula = tseitin_formula(ring)
        length = resolution_length(formula)

        total_resolution_length += length

        if length < 2 ** min_rank:
            conjecture_holds = False
            counterexample = f"Ring of size {n} with rank {min_rank} and proof length {length}"

metric_value = total_resolution_length / instances_tested

return {
    "metric_name": "Resolution Proof Length",
    "metric_value": metric_value,
    "instances_tested": instances_tested,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(seed) for seed in sys.argv[1:]] if sys.argv[1:] else [2**i + 3 for i in range(5,

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

```

```

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
else:
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b3a3f6f.py", line 84, in <module>
    result = run_trial(seed)
              ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_3b3a3f6f.py", line 51, in run_trial
    boolean_rings = [random.sample([0, 1], n) for _ in range(5)]
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 430, in sample
    raise ValueError("Sample larger than population or is negative")
ValueError: Sample larger than population or is negative

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing any data, which prevents us from evaluating whether the mean resolution proof length meets the required condition. | next: Re-run the test ensuring it completes successfully and produces the necessary data to evaluate the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	15675
2	preregistration	ollama_remote	glm4:latest	0	0	9754
3	novelty	ollama_remote	glm4:latest	0	0	9195
4	novelty	ollama_remote	glm4:latest	0	0	9269
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	41439
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10072
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10893
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8674
9	judge	ollama_remote	glm4:latest	0	0	14894

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 129863 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/c87e07b4e9c9.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/c87e07b4e9c9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/c87e07b4e9c9.tar.gz` (if generated)